ICT2142 E-Business Systems

Lecture 10

Emerging Trends in E-Business



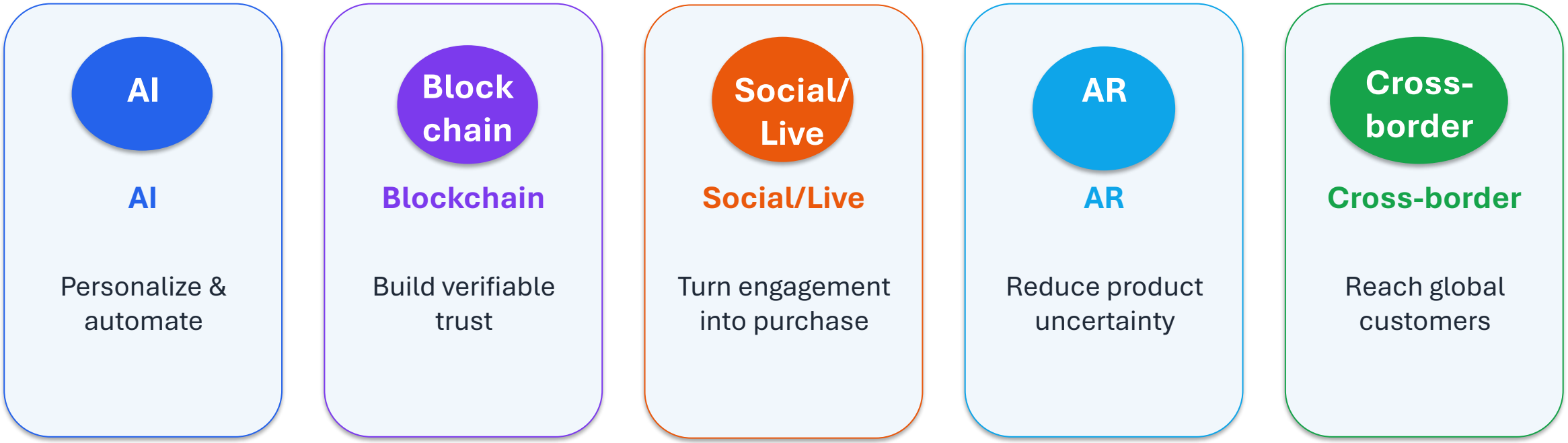
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Learning Outcomes

- Use of AI in E-Commerce.
- Blockchain and smart contracts
- Social commerce and live commerce
- Augmented reality (AR) in online shopping
- Cross-border e-commerce

Why These Trends Matter

E-business is moving from static online storefronts toward intelligent, immersive, automated and borderless commerce.

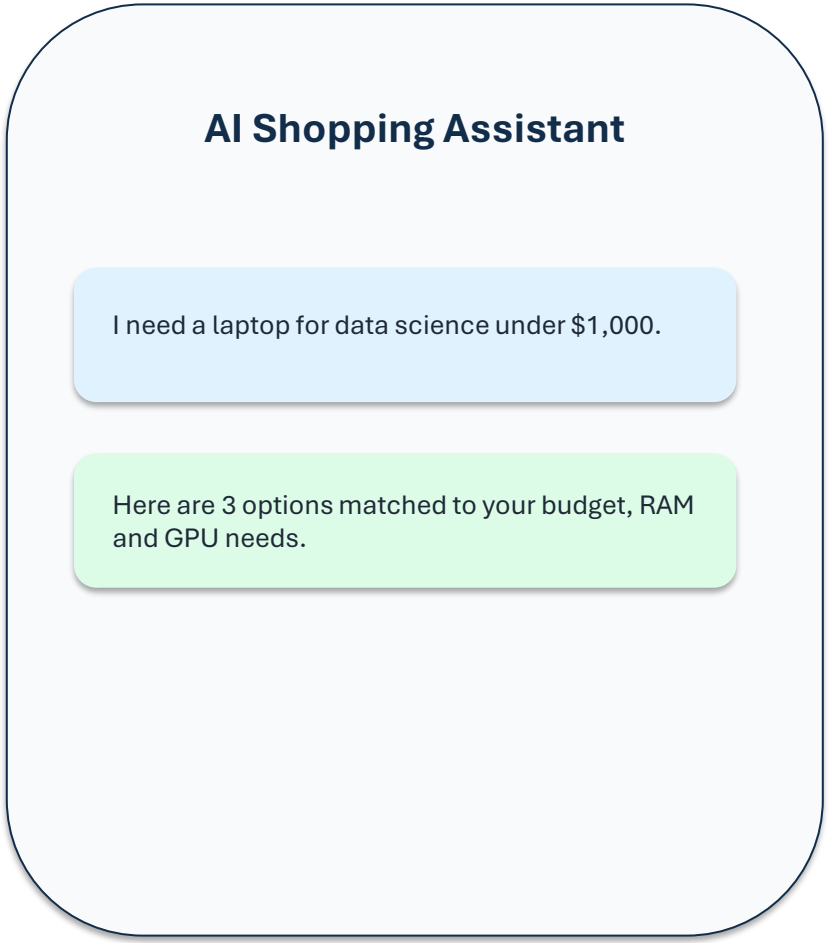


Core idea: technology creates value only when it improves customer experience, operational efficiency, trust, market reach, or decision quality.

AI in E-Commerce

Artificial intelligence uses data and algorithms to support prediction, personalization, automation and natural-language interaction.

- Recommendation engines suggest products based on user behavior, product similarity, context and historical interactions.
- Chatbots and AI shopping assistants answer questions, compare products, provide support and guide customers toward checkout.
- AI can also support demand forecasting, fraud detection, dynamic merchandising and customer segmentation.
- Value comes from better relevance and faster service; risks include privacy, bias, inaccurate recommendations and over-automation.



AI Recommendation Engines

A recommender estimates which products are most relevant to a particular shopper or context.

1

Collect signals

Clicks, searches, purchases, ratings, cart activity

2

Build representation

User preferences + product attributes

3

Generate candidates

Collaborative, content-based or hybrid methods

4

Rank

Score items by predicted relevance

5

Recommend

Display personalized products and learn from feedback

Example: A shopper who repeatedly views running shoes and sports watches may receive recommendations for related footwear, fitness wearables and accessories.

Common approaches: collaborative filtering • content-based filtering • hybrid recommenders • deep-learning/LLM-assisted systems

AI Chatbots and Conversational Commerce

Chatbots provide an immediate conversational interface for product discovery, support and transaction-related questions.

Before purchase

Product questions
Recommendations
Comparison
Availability

During purchase

Sizing/compatibility
Checkout guidance
Promotions
Cart assistance

After purchase

Order status
Returns guidance
FAQs
Human-agent handoff

Traditional chatbot

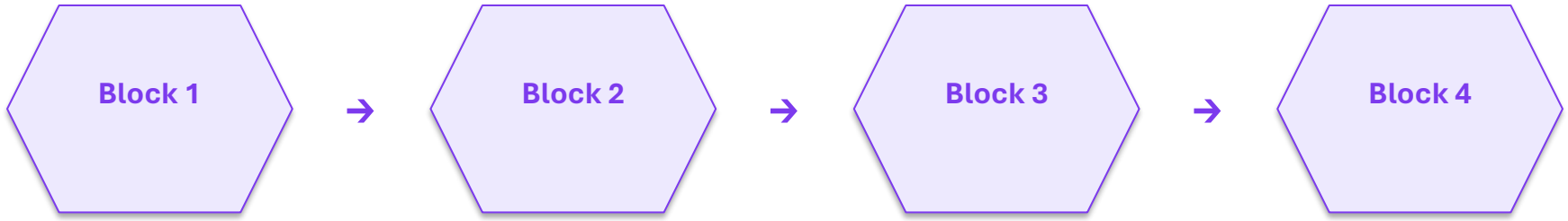
Rule-based flows and predefined answers. Reliable for narrow FAQs but limited when questions vary.

AI / generative assistant

Uses natural language and store knowledge to handle flexible questions; requires grounding, guardrails and monitoring.

Blockchain in E-Business

A blockchain is a distributed ledger in which validated transactions are recorded across a network, creating a tamper-evident history.



- Shared record across participating nodes.
- Cryptographic linking helps reveal unauthorized alteration.
- Can improve traceability and auditability in multi-party processes.
- Useful where organizations need a common record without relying on one database owner.

E-business applications

Supply-chain provenance
Digital identity and credentials
Tokenized assets
Cross-organizational records
Automated agreements through smart contracts

Smart Contracts

Smart contracts are programs stored on a blockchain that execute predefined actions when specified conditions are satisfied.



Benefits

- Automation of agreed rules
- Transparent execution history
- Reduced manual reconciliation
- Faster multi-party workflows

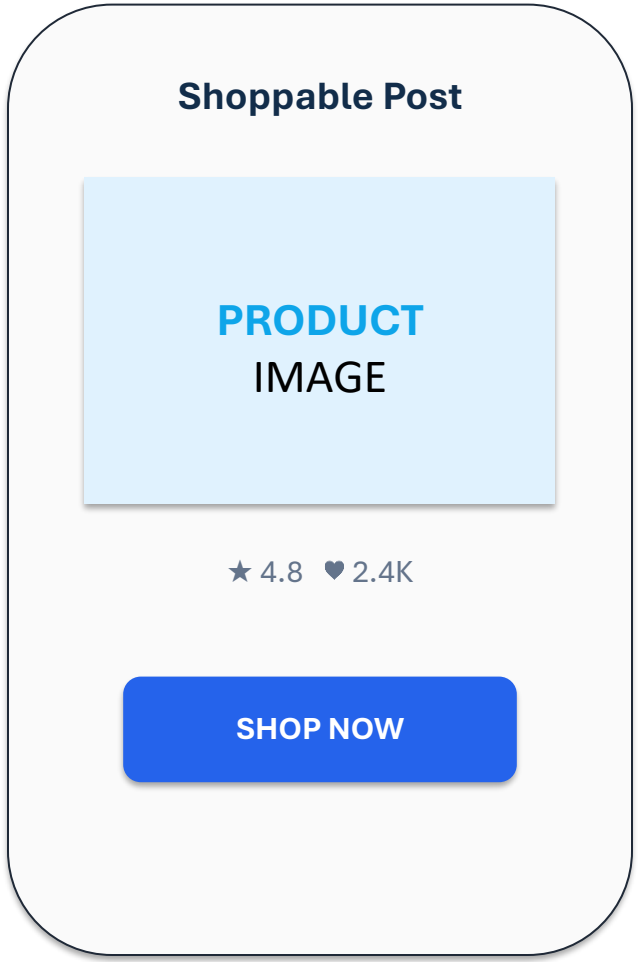
Limitations / risks

- Code defects can automate the wrong outcome
- External data ('oracles') may be unreliable
- Legal interpretation and governance remain important
- Public chains may raise privacy/scalability concerns

Social Commerce

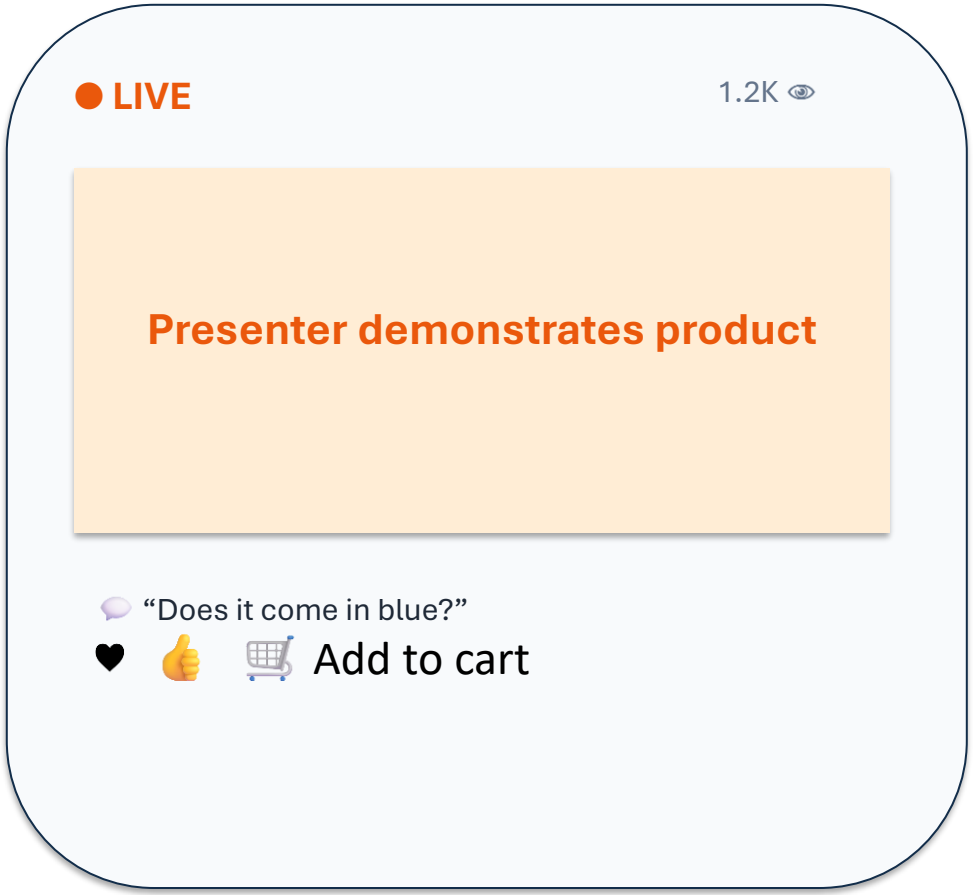
Social commerce integrates product discovery, interaction and purchasing into social-media or community experiences.

- Customers discover products through posts, creators, communities, short-form video and recommendations.
- Shoppable content reduces the distance between inspiration and purchase.
- Social proof - comments, reviews, shares and creator demonstrations - can influence trust and product discovery.
- Businesses can connect product catalogs to social channels and track engagement-to-purchase journeys.
- Risks include platform dependence, misleading promotion, privacy concerns and reputational damage.



Live Commerce

Live commerce combines real-time video, product demonstration, audience interaction and purchasing.

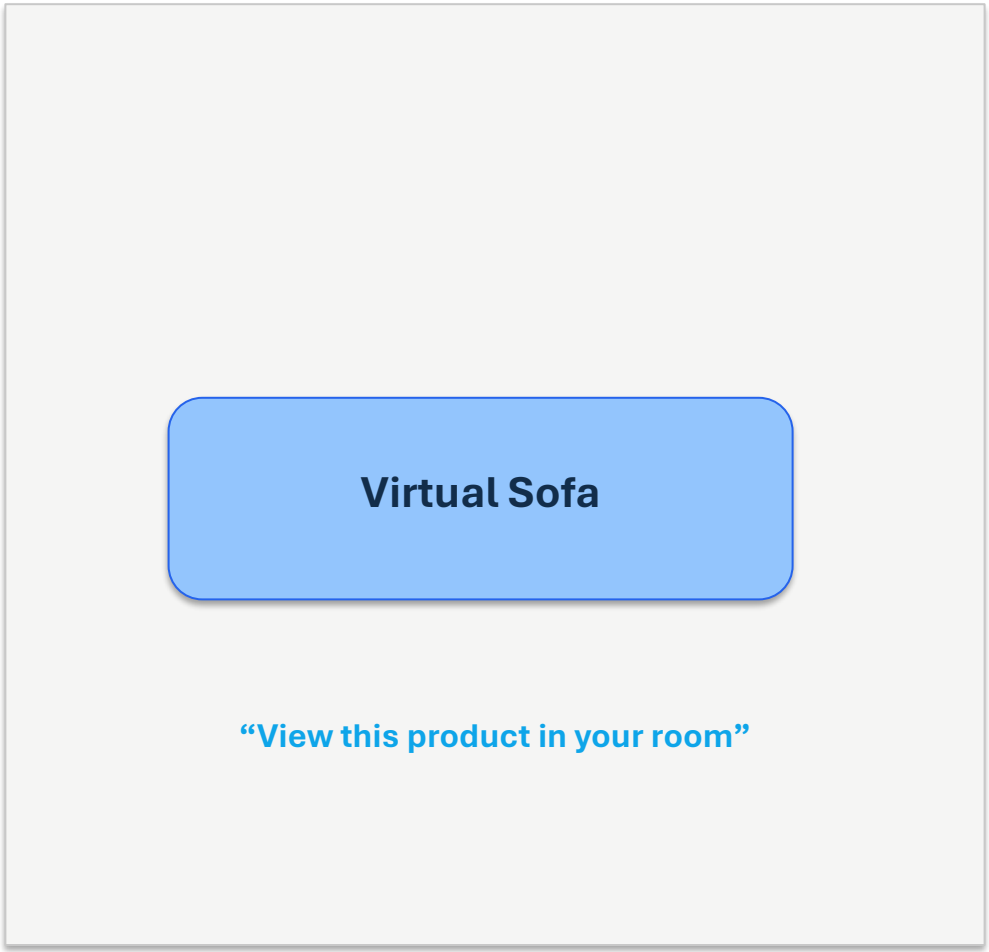


- Host demonstrates products and answers questions in real time.
- Viewers can react, comment and purchase without leaving the experience.
- Creates urgency through limited offers, launches or timed promotions.
- Useful for fashion, beauty, electronics, collectibles and product demonstrations.
- Success depends on credible hosts, reliable streaming, inventory synchronization and transparent promotion.

Augmented Reality (AR) in Online Shopping

AR overlays digital content or 3D product models onto the shopper's real-world view, commonly through a smartphone camera.

- Virtual try-on: eyewear, cosmetics, clothing and accessories.
- View-in-room: furniture, décor and appliances at realistic scale.
- 3D product inspection: rotate, zoom and examine details before purchase.
- Business value: improves product understanding and buying confidence; may reduce uncertainty and returns.
- Requirements: accurate 3D assets, device/browser support, good UX and responsible handling of camera-related data.



Cross-Border E-Commerce

Cross-border e-commerce occurs when buyers and sellers in different countries conduct digitally enabled commercial transactions.



- Opportunities: larger customer base, niche global markets, digital services and international marketplace access.
- Localization: language, currency, product information, customer support and culturally appropriate marketing.
- Payments: exchange rates, payment-method preferences, fraud controls and transaction costs.
- Logistics: shipping cost, delivery time, tracking, returns and carrier selection.
- Compliance: customs, duties/taxes, product rules, consumer protection, privacy and intellectual-property requirements.

Comparing the Emerging Trends

Each trend solves a different e-business problem; many organizations combine them.

Trend	Primary value	Typical use	Key risk
AI	Personalization & automation	Recommendations, chatbots	Bias, privacy, inaccurate output
Blockchain	Shared trust & traceability	Provenance, multi-party records	Scalability, governance
Social / Live	Discovery & engagement	Shoppable content, streams	Platform/reputation risk
AR	Product visualization	Virtual try-on, view-in-room	Cost, device/data concerns
Cross-border	Market expansion	International selling	Customs, payments, logistics

Integrated E-Business Scenario

A Sri Lankan fashion retailer wants to expand online sales locally and internationally.



AI

Recommend products based on browsing and answer sizing/availability questions.



Social + Live

Launch products through shoppable short videos and live demonstrations.



AR

Let customers preview selected accessories or apparel digitally.



Blockchain

Use where justified for provenance or shared supply-chain records.



Cross-border

Localize currency/content and manage payments, shipping, customs and returns.

Thank You...